What is an RC (ReplicationController) in Kubernetes?

A **ReplicationController (RC)** is a legacy Kubernetes object used to ensure that a specified number of **pod replicas** are running at any given time.

It watches pods and restarts or creates them if they fail or are deleted, ensuring **high availability**.

Key Responsibilities of RC:

- Maintains a stable set of replica pods.
- Ensures the desired number of pods are running.
- Replaces failed pods automatically.

⚠️ **Note:** In modern Kubernetes, **Deployments** are preferred over ReplicationControllers because they offer more features (like rolling updates, rollbacks, etc.).

Example: ReplicationController YAML

```
apiVersion: v1
kind: ReplicationController
metadata:
  name: nginx-rc
spec:
  replicas: 3
  selector:
    app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx-container
        image: nginx:latest
        ports:
        - containerPort: 80
```

Explanation:

- **replicas**: Ensures 3 pods are always running.
- **selector**: Matches the labels in pod templates.
- **template**: Defines the pod configuration.

How to Create RC:

```
kubectl apply -f nginx-rc.yaml
```

Check the RC and Pods:

```
kubectl get rc
kubectl get pods
```